

Sam Goodwin

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Profile

Computer Science graduate (Bridgewater State, 2025, Magna Cum Laude) with hands on experience across backend development, concurrent systems, computer vision, and IoT. Comfortable working in Golang, Java, Python, and C++, and building software that runs cleanly across platforms. Passionate about well-structured, well-documented code and picking the right tool for the problem.

Education

Bachelor of Science in Computer Science

Bridgewater State University, 2025

- Magna Cum Laude (GPA: 3.8)
- Dean's List (All Semesters)

Skills

Languages: Golang, Python, Java, C++, Bash/Shell

Databases: SQL and SQLite

Libraries/Frameworks: OpenCV, DJI Tello, Excelize, SQLite3, Fyne, JavaFX, Maven, face_recognition, MediaPipe

Tools: Git, GitHub, Arduino, Docker, Raspberry Pi, VS Code, PlatformIO, JetBrains

Operating Systems: Linux and Windows

Other: GUI/CLI development, Computer vision, Data Visualization, Mapping, IoT Systems, API integration

Certifications

- SQL (Advanced, Intermediate, Basic) Skills Certification – HackerRank
- Go (Intermediate and Basic) Skills Certification – HackerRank
- Java (Basic) Skills Certification – HackerRank
- Python (Basic) Skills Certification – HackerRank

Projects

Job Visualizer

GitHub: github.com/goodwin-sam/job-visualizer

- Cross-platform Go desktop app (GUI + CLI) for importing, processing, and visualizing job data from Excel files, backed by a SQLite database.
- Built interactive GUI with Fyne for file filtering by keyword, location, salary, and remote status, plus browser-based mapping via OpenStreetMap Nominatim API with local geocoding cache, and interactive charts for data visualization.
- Maintained clean architecture through cmd/pkg separation, conventional commits, comprehensive testing, detailed documentation, and portable release binaries for Linux and Windows.

Concurrent File Processor

GitHub: github.com/goodwin-sam/concurrent-file-processor

- Cross-platform Java desktop app for concurrent multi-file text analysis, computing word frequencies, character/line counts, and aggregate file statistics via GUI or headless CLI.
- Implemented custom work-stealing thread pools for load-balanced parallel processing, with a JavaFX multi-window GUI and portable executables for Windows and Linux.
- Designed thread-safe modular architecture with clear package separation, comprehensive unit tests, and thorough error handling.

Smart Tello Follower

GitHub: github.com/goodwin-sam/smart-tello-follower

- Python computer vision system that streams live video from a DJI Tello drone, performs real-time face recognition using OpenCV and the face_recognition library to autonomously track a target.
- Implemented gesture recognition via MediaPipe to allow hand-signal-based flight control, with modular separation of drone control, face recognition, and gesture recognition into distinct components.
- Designed configurable settings via a dedicated config module, with clean dependency management.

Environment Sensor Hub

GitHub: github.com/goodwin-sam/enviro-sensor-hub

- Arduino-based C++ environmental monitoring system that reads temperature, humidity, water level, and ambient light from multiple sensors, driving a fan, buzzer, and 16x2 LCD in response.
- Implemented PWM fan speed control across three temperature tiers, automatic LCD backlight adjustment based on ambient light, and configurable buzzer alarms for threshold violations.
- Organized into a modular package structure (sensors, display, serial utilities) with all thresholds and pin assignments centralized in a config header for easy tuning.

Experience

Property Manager

Private Landlord, Massachusetts

Summer 2020 – Present

- Managed day to day operations for multiple residential rental properties, responding to landlord directed tenant requests often same day or next day. Including on-call night/weekend work for critical issues such as no heat or no hot water.
- Performed hands on handyman repairs, lawn care, snow removal and general maintenance personally, troubleshooting and/or resolving plumbing, electrical, appliance, and other general issues.
- Collected and processed rental payments while maintaining accurate records and handling administrative tasks.
- Scheduled and oversaw external contractors when needed, ensuring timely completion and quality work.
- Handled documentation, scheduling, and follow-ups, demonstrating strong organization and reliability in a fast-paced, high demand environment.

Teaching Assistant/Tutor

Curry College, Milton Massachusetts

January 2019 – May 2020

- Delivered one-on-one and small-group tutoring sessions for undergraduate students in Chemistry and Biology courses.
- Assisted students in understanding complex scientific material, breaking down problems, explaining experimental processes, and improving critical thinking and problem-solving skills.
- Held regular office hours and review sessions to fix conceptual misunderstandings, review homework/lab reports, and provide constructive feedback to enhance comprehension and academic performance.
- Supported faculty by preparing study materials, facilitating group discussions, and contributing to student success in demanding science curriculum.